

QA PPT New 60 hrs (1)

Find the largest number which can divide x, y, z exactly.	HCF of (x, y, z)
Find the largest number which can be divided by x, y, z and leaves same remainder in each case.	HCF of $[x-y, y-z, z-x]$
Find the largest number which can divide x, y, z and leaves remainder a, b, c respectively.	HCF of $[(x-a), (y-b), (z-c)]$

10. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case:

A.4 B.7 C.9 D.13

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10. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case:

A.4 B.7 C.9 D.13

$$\text{HCF} (193-91, 191-183, 183-43)$$

$$\text{HCF} (48, 92, 140)$$

$44 \rightarrow 1, 2, 4, 11, 22, 44$

13. A rectangular field of dimension 180m x 105m is to be paved by identical square tiles. Find the size of each tile and the number of tiles required.

A.60 B.78 C.84 D.90

$$\begin{aligned} & [180, 105] \\ & \text{HCF} = 15 \\ & \text{Size of tile} = 15 \times 15 \\ & \text{No. of tiles required} = \frac{180 \times 105}{15 \times 15} \end{aligned}$$

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Product of two numbers = LCM X HCF

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Concept of Unit Digit:-

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457 ~~x~~ 358 ~~x~~ 574

6 4

872 ~~x~~ 758 ~~x~~ 346 ~~x~~ 432

6 2 2

765 + 397 ~~x~~ 239 - 358

9 3 0 - - - - -

976 ~~x~~ 567 ~~x~~ 348 ~~x~~ 873

2 4 8

322 ~~x~~ 498 + 654 - 397

6 10 3

521 876 567 987

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➤ Cyclicity of 2, 3, 7, 8 – 4
 ➤ Cyclicity of 4, 9 – 2
 ➤ Cyclicity of 0, 1, 5, 6 – 1

98/4 R=2
 2
 2²
 4

38/4 R=2
 4
 4²
 6

74/4 R=2
 7
 7²
 9

60/4 R=0
 2
 2⁴
 6

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Units place of any no greater than 5! Will always be '0' (since we will mul by 5 & 2 in each)

Q] units place of $1! + 4! + 999! + 10000! + 1212!$ is

$$4! = 24 \rightarrow \text{UP} = 4$$

$$2! = 2 \rightarrow \text{UP} = 2$$

UP of rem nos $> 5!$ Hence they are 0

$$\text{Therefore } 4 + 2 + 0 = 6$$

Hence the UP of above expression will be 6

Q] UP of

$2! \times 3! \times 4! + 5! \times \dots \times 100!$ Is

UP of $2! = 2$

Of $3 = 6$

Of $4 = 4$

All above $5! = 0$

Hence UP for above exp will be

$2 \times 6 \times 4 + 0 = 58$

Therefore UP for above exp is 8



Factor Theorem:-

➤ How can we find out no. of factors of any given number?

$$12 = 1, 2, 3, 4, 6, 12$$

$$12 = 2 \times 2 \times 3$$

$$\underline{2^2} \times \underline{3^1}$$

$$\text{No. of factor}(n) = (2+1)(1+1)$$

$$= 6$$

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➤ How can we find out product of factors of any given number?

$$12 = 1, 2, 3, 4, 6, 12$$

$$12 = 2 \times 2 \times 3$$

$$= \underline{2^2} \times \underline{3^1}$$

$$P.O.F = 12^{6/2}$$

$$= 12^3$$

$$P.O.F = (N)^{n/2}$$

$N \rightarrow$ given no.
 $n \rightarrow$ Total no. of factor
 of given no.

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$$\frac{x+a}{x} \Rightarrow \frac{x}{x} + \frac{a}{x} \Rightarrow R=a$$

$$\frac{(x+a)^2}{x} \Rightarrow \frac{x^2}{x} + \frac{2ax}{x} + \frac{a^2}{x} \Rightarrow R=a^2$$

$$\frac{(x+a)^3}{x} \Rightarrow \dots \Rightarrow R=a^3$$

$$\vdots$$

$$\frac{(x+a)^n}{x} \Rightarrow \dots \Rightarrow R=a^n$$

$$\frac{(x-a)^n}{x} \Rightarrow \dots \Rightarrow R=(-a)^n$$

$$\left. \begin{array}{l} \text{nieven} \\ R=a^n \end{array} \right\} \quad \left. \begin{array}{l} \text{ni odd} \\ R=-a^n \end{array} \right\}$$

$$\frac{(x+1)^n}{x} \rightarrow R=1$$

$$\frac{(x-1)^n}{x} \rightarrow R=(-1)^n$$

$$\begin{array}{l} \text{nieven} \quad \text{ni odd} \\ R=1 \quad R=-1 \end{array}$$

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remainder?

A. 2

B. 4

C. 1

D. 3

$$\begin{array}{r} 96 \\ 2 \overline{) 192} \\ \underline{184} \\ 8 \end{array} \Rightarrow \frac{(2^3)^{32}}{7}$$

$$\frac{8^{32}}{7}$$

$$\frac{(7+1)^{32}}{7} \Rightarrow \boxed{R=1}$$

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Fermat's Theorem:

$$\frac{a^{p-1}}{p} \quad R=1$$

$p \rightarrow$ Prime no.

$$\frac{2^4}{5} \text{ or } \frac{16}{5} \quad R=1$$

$$\frac{2^{5-1}}{5} \quad R=1$$

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Wilson's Theorem:

$$\frac{(p-1)!}{p} \quad R=p-1$$

$p \rightarrow$ Prime no

$$\frac{4!}{5} \quad R=4$$

$$\frac{(5-1)!}{5} \quad R=5-1 \quad R=4$$

